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ORIGINAL ARTICLE

Attenuation of *Pseudomonas aeruginosa* biofilm by thymoquinone: an individual and combinatorial study with tetrazine-capped silver nanoparticles and tryptophan

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Abstract

Microbial biofilm indicates a cluster of microorganisms having the capability to display drug resistance property, thereby increasing its proficiency in spreading diseases. In the present study, the antibiofilm potential of thymoquinone, a black seed–producing natural molecule, was contemplated against the biofilm formation by *Pseudomonas aeruginosa*. Substantial antimicrobial activity was exhibited by thymoquinone against the test organism wherein the minimum inhibitory concentration of the compound was found to be 20 μg/mL. Thereafter, an array of experiments (crystal violet staining, protein count, and microscopic observation, etc.) were carried out by considering the sub-MIC doses of thymoquinone (5 and 10 μg/mL), each of which confirmed the biofilm attenuating capacity of thymoquinone. However, these concentrations did not show any antimicrobial activity. Further explorations on understanding the underlying mechanism of the same revealed that thymoquinone accumulated reactive oxygen species (ROS) and also inhibited the expression of the quorum sensing gene (*lasI*) in *Pseudomonas aeruginosa*. Furthermore, by taking up a combinatorial approach with two other reported antibiofilm agents (tetrazine-capped silver nanoparticles and tryptophan), the antibiofilm efficiency of thymoquinone was expanded. In this regard, the highest antibiofilm activity was observed when thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles were applied together against *Pseudomonas aeruginosa*. These combinatorial applications of antibiofilm molecules were found to accumulate ROS in cells that resulted in the inhibition of biofilm formation. Thus, the combinatorial study of these antibiofilm molecules could be applied to control biofilm threats as the tested antibiofilm molecules alone or in combinations showed negligible or very little cytotoxicity.

Introduction

Bacteria commonly exhibit two forms of growth in nature: one is planktonic while the other one is known as biofilm (Gebreyohannes et al. 2019). Planktonic bacteria are single-celled microbes that prefer to live in discrete mode and remain separated from each other (Sretenovic et al. 2017). On the contrary, the biofilm is a 3D structure, formed when the microbial cells aggregate on biotic or abiotic surfaces through different non-covalent interactions including van der Waals

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forces and hydrophobic interactions (Gupta et al. 2016). Both homogenous and heterogeneous microbial populations have been reported to develop biofilm over different surfaces (Gupta et al. 2016). Bacteria stabilize the structure of biofilm by secreting a self-producing matrix commonly known as extracellular polymeric substances (EPS). EPS mostly consists of polysaccharides but other biomolecules like proteins, lipids, and nucleic acids are also spotted in this matrix during the development of microbial biofilm (Di Martino 2018). EPS layer often impedes the smooth propagation of antimicrobials through the biofilm, thus making it resistant to these agents (Di Martino 2018). Moreover, biofilm has also been found to be resistant against bacteriophages, synthetic biocides, amoebae, etc. (Gupta et al. 2016). It was reported that biofilm cells could show drug resistance up to 1000-fold in contrast to the drug resistance property exhibited by the planktonic cells of the same organism (Gebreyohannes et al. 2019). Besides, microbial biofilm was reported to cause infections in the host by skipping the host immunity (Moser et al. 2017). It was also stated that bacterial biofilm was found to cause various chronic and life-threatening diseases in humans (Maurice et al. 2018). As per the report of the Centre for Disease Control (CDC) and the National Institute of Health (NIH), approximately 65% and 80% of the chronic microbial infections have been associated with microbial biofilm, respectively (Jamal et al. 2019).

Pseudomonas aeruginosa happens to be a Gram-negative rod-shaped bacterium that has been found to develop biofilm in human hosts under various environmental conditions (Diggle and Whiteley 2020). *Pseudomonas aeruginosa* has been found to develop biofilm in joint prostheses, heart valves, urinary catheters, endocarditis, periodontitis, osteomyelitis, rhino-sinusitis, cystic fibrosis, and kidney diseases (Gupta et al. 2016). Thus, novel strategies are to be adopted rapidly to inhibit the formation of microbial biofilm. In this regard, small molecules have been proven to act effectively against the development of microbial biofilm (Ghosh et al. 2020). It has been documented that thymoquinone, a phytochemical compound present in the seed oil of dark cumin, displayed a promising antimicrobial activity against *Pseudomonas aeruginosa* (Yimer et al. 2019). In the present report, we have put our efforts to investigate the effect of thymoquinone on the inhibition of biofilm development by *Pseudomonas aeruginosa*. Efforts have also been given to understand the mechanism of microbial biofilm inhibition under the influence of the tested antibiofilm molecules. The result of the current study demonstrates that thymoquinone significantly inhibits the microbial biofilm formation by accumulating reactive oxygen species (ROS) and interfering with quorum sensing. However, existing literature revealed that the combinatorial approach was found to work more effectively than individual ones in the context of microbial biofilm inhibition (Das et al. 2016a; Tagliaferri et al. 2019). In this connection, it was reported that combinations of vitexin, azithromycin, and gentamicin had shown substantial inhibition in the biofilm development by *Pseudomonas aeruginosa* (Das et al. 2016b). In a separate report, it was documented that combinations of plumbagin and gentamicin could be applied effectively to attenuate the biofilm formation by *Pseudomonas aeruginosa* (Gupta et al. 2017). Moreover, it was also reported that patients suffering from Pseudomonal infections taking both antipseudomonal β-lactam and aminoglycoside antibiotics were found with a higher cure rate than the patients taking only aminoglycoside antibiotic (Voulgaris et al. 2019). Thus, to increase the antibiofilm activity of thymoquinone, in the current report, a combinatorial study has been carried out with two others reported antibiofilm agents namely tetrazine-capped silver nanoparticles (TzAgNPs) (Chakraborty et al. 2018a) and tryptophan (Chakraborty et al. 2018b) against the biofilm development by *Pseudomonas aeruginosa*. The present study reveals that the combination of all the antibiofilm compounds work effectively against the biofilm development in contrast to their individual influence.

Materials and methods

Microbial strain and culture conditions

Pseudomonas aeruginosa (accession number MTCC 424) was used as a test organism for the present study. Dr. Surajit Bhattacharjee (Tripura University) had given the mentioned organism to us only for the research work purpose. Luria broth (LB) medium (Himedia, India) was used as the preferred medium for the optimum growth of the said organism. The organism was incubated at 37 °C for a varied length of time as per the requirement of the experiment. Thymoquinone (purity ≥ 98%) used in the current study was purchased from Sigma-Aldrich (274666).

Agar diffusion assay

To analyze the antibacterial property of thymoquinone against *Pseudomonas aeruginosa*, agar diffusion assay was conducted by following the procedure as reported previously (Bauer et al. 1996; NCCLS 1997). In brief, the cells of *Pseudomonas aeruginosa* (1×10^7 colony-forming unit (CFU/mL) were mixed with the sterile LB agar media at permissible temperature and finally poured on sterile Petri dishes to prepare the microbial lawn over it. Thereafter, different concentrations of thymoquinone were added separately to the wells punctured on the plates. All the plates were then incubated at 37 °C for 48 h. The extent of growth inhibition caused by thymoquinone was determined by measuring the diameter of the clear zones formed around the wells.

Determination of minimum inhibitory concentration and minimum bactericidal concentration of thymoquinone

To determine the minimum inhibitory concentration (MIC) of thymoquinone against *Pseudomonas aeruginosa*, the broth dilution system (BDS) was followed (Chakraborty et al. 2018a). To do so, a similar number of the cells of *Pseudomonas aeruginosa* (1×10^7 CFU/mL) were separately added to different tubes filled with 5 mL of autoclaved LB media. They were further challenged with various concentrations (5, 10, 20, and 30 μg/mL) of the compound (thymoquinone). However, a control set was kept in which cells were grown without being exposed to thymoquinone. Thereafter, all the growth media were allowed to grow for 48 h at 37 °C. After that, the turbidity of the culture media collected from each growth media was examined by measuring its optical density (OD) at 600 nm. Thereafter, to determine the minimum bactericidal concentration (MBC) of thymoquinone, a similar number of bacterial cells (1×10^7 CFU/mL) were inoculated into different test tubes filled with 5 mL of sterile LB growth media. After that, an increased concentration of thymoquinone was separately added to the growth media and incubated for 48 h at 37 °C. After the desired period of incubation got over, the microbial viability was measured in each test tube by determining the CFU by following the protocol described by Chakraborty et al. (2018b).

Determination of microbial biofilm formation by crystal violet assay

To examine the role of thymoquinone on the biofilm-forming ability of *Pseudomonas aeruginosa*, crystal violet (CV) assay was performed by following the method described previously (Mukherjee et al. 2013; Sharma et al. 2015). In brief, a similar number of *Pseudomonas aeruginosa* (1×10^7 CFU/mL) were separately added into different tubes carrying 5 mL of sterile LB growth media. After that, different concentrations (5 and 10 μg/mL) of thymoquinone were added to the growth media. In some experiments, a similar number of microbial cells were separately exposed to various combinations of the tested antibiofilm molecule. However, in the control set, the organisms were allowed to grow in the absence of any of the tested antibiofilm molecule. Thereafter, all the growth media including control were then incubated at 37 °C for 48 h. Following the incubation period, all the free-living non-adhered cells were discarded from each tube and aseptically washed off with the help of autoclaved Milli-Q water. Then, the test tubes were air-dried and stained with CV solution (0.4%) for 15 min to determine the adhered microbial population on the glass surfaces. Thereafter, the CV solution was carefully removed from each tube, gently washed with Milli-Q water, and air-dried. To dissolve the adsorbed CV by the microorganisms adhered to the glass surface, glacial acetic acid (33%) was added to each tube. Finally, the amount of the colour generated in all the tubes was determined by measuring the absorbance at a wavelength of 630 nm.

Microbial colonization analysis under fluorescence microscopy

To test the degree of bacterial colonization on the glass surfaces under the influence of thymoquinone, a nearly equal number of *Pseudomonas aeruginosa* (1×10^7 CFU/mL) was separately inoculated into various sterile tubes carrying 5 mL of autoclaved LB growth media. Then, the test concentrations (5 and 10 μg/mL) of thymoquinone were carefully added to the respective tubes. In some experimental sets, various combinations of the tested antibiofilm molecules were added accordingly. A control set was marked where a similar number of microorganisms were grown in absence of any of the mentioned antibiofilm agents. After that, autoclaved glass coverslips were diligently incorporated into all the growth media and subsequently incubated at 37 °C for 48 h. Post incubation, the coverslips were aseptically recovered from every conditioned media. After that, coverslips were washed with sterile double-distilled water to remove unattached bacteria. Thereafter, the coverslips were air-dried, stained with acridine orange (4 μg/mL) solution, and observed under a fluorescence microscope.

Measurement of total biofilm protein

The extraction and measurement of total biofilm protein gives an indirect measurement of the degree of microbial colonization to the glass surface as the protein gets originated from the microorganisms only (Das et al. 2016a; Das et al. 2016b). To measure the degree of microbial biofilm formation on glass surface under the influence of thymoquinone, cells were inoculated in LB and further challenged with various sub-MIC concentrations (5 and 10 μg/mL) of thymoquinone. In the control set, cells were grown without being exposed to thymoquinone. All the growth media were then incubated at 37 °C for 48 h. Followed by the incubation, the free-flowing planktonic cells were removed from all the test tubes. Then, these tubes were re-washed with sterile Milli-Q water and subsequently air-dried. These tubes were then boiled in 5 mL of 0.3 (mol/L) NaOH for 30 min. The boiled suspension was subsequently centrifuged for 10 min at 3000×*g*. Protein concentration in the supernatant was measured by the Lowry method (Lowry et al. 1951).

Quantification of the extracellular polymeric substances

EPS released by the test organism (*Pseudomonas aeruginosa*) under the influence of thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles individually or in combinations were determined by adhering to the protocol described by Tribedi and Sil (2014). To quantify the same, 100 μL culture of *Pseudomonas aeruginosa* (1×10^7 CFU/mL) were independently inoculated in various plates (60 mm sterile polystyrene plate) containing sterile LB growth media treated with various combinations of antibiofilm molecules (tryptophan, tetrazine-capped silver nanoparticles, and thymoquinone). However, for the control set, a similar number of microorganisms were grown in LB without being exposed to the mentioned antibiofilm molecules. Each growth media was then incubated for 2 days at 37 °C following which the planktonic cells were removed. Biofilm formed at the bottom of the polystyrene plate was separately collected with the help of a sterile scrapper by putting sterile Milli-Q water. The biofilm suspension was then centrifuged at 3500×*g* for 20 min at 4 °C to collect the supernatant in which EPS accumulated. Next, the cell pellet was collected and further treated with 10 mM EDTA and re-centrifuged for the extraction of the cell-bound EPS (if any). The supernatant was recovered and added to the previously collected supernatant. The pooled supernatant was mixed with chilled absolute ethanol (2.2 volume of pooled supernatants). This mixed suspension was then incubated for the next 1 h at 20 °C and further centrifuged for 20 min at 3000×*g* at 4 °C. Finally, the phenol-sulphuric acid method was followed to quantify the extent of EPS produced by *Pseudomonas aeruginosa* under different conditions (Dubois et al. 1956).

Intracellular measurement of reactive oxygen species

To measure the extent of ROS accumulation in microorganisms under the influence of thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles individually or in combinations, *Pseudomonas aeruginosa* were allowed to grow in different tubes filled with LB (5 mL) until they reached 1×10^8 CFU/mL. Thereafter, DCFDA was given to the microbial culture in all experimental sets at a ratio of 1:2000 and shaken for 30 min at a temperature of 37 °C. Post incubation, the microbial culture was centrifuged for 10 min at 3500×*g* to settle down bacterial cells and further washed to remove the unabsorbed DCFDA. Thereafter, the bacterial cells were exposed to various combinations of the mentioned antibiofilm agents. In some experimental set, ascorbic acid (a strong antioxidizing agent) was added immediately after the exposure of different combinations of the antibiofilm molecules (tryptophan, tetrazine-capped silver nanoparticles, and thymoquinone) to the cells. A control set was also kept wherein a similar number of the organisms were allowed to grow neither in the presence of any of the mentioned antibiofilm molecules nor ascorbic acid. The fluorescence intensity of the generated DCF was then determined by a fluorescence spectrophotometer at an excitation and emission wavelength of 488 nm and 535 nm, respectively (Dwivedi et al. 2014).

Effect of thymoquinone on the genomic DNA integrity of *Pseudomonas aeruginosa*

To study the effect of thymoquinone on the genomic DNA integrity of *Pseudomonas aeruginosa*, a similar number of organisms (1×10^8 CFU/mL) were treated with different concentrations (5 and 10 μg/mL) of thymoquinone and subsequently incubated at 37 °C for 48 h. In the control set, an equal number of microbial cells were grown in the absence of thymoquinone. Thereafter, each growth media including control was separately centrifuged for 5 min at 3500×*g* at refrigerated condition (4 °C). The cell pellet and the supernatant were separately collected from each growth media. The DNA was subsequently isolated from each recovered cell pellet by following the protocol described by Singh (2016). Thereafter, the isolated DNA from the cell pellet and the supernatant of each growth media was subjected to agarose gel (0.9%) electrophoresis and subsequently visualized by gel documentation system (Vilber Lourmat model no. ECX 20.M).

Swarming motility assay

Pseudomonas aeruginosa show a distinct mode of motility (swarming motility) with the help of their flagella. To examine the swarming motility of the organism under different conditions, 100 mL of LB was separately autoclaved in three conical flasks in which 0.6% agar and 0.5% glucose were also added. In the first conical, the sterile media was mixed with 5 μg/mL of thymoquinone. In the second one, 10 μg/mL of thymoquinone was added to the autoclaved media. However, in the last one, the media was not mixed with thymoquinone. Thereafter, media from each conical was separately poured into different sterile Petri dishes. After the agar solidified in all plates, a similar number of microbial cells of *Pseudomonas aeruginosa* (1×10^7 CFU/mL) was separately spotted in the middle of the agar plates. All the plates were then incubated at 37 °C for 2 days. Post incubation, the microbial colony diameter was measured in all the plates.

Quorum sensing gene expression measurement by real-time PCR

Real-time PCR was used to measure the expression of a target DNA molecule under different conditions (Chakraborty et al. 2018b). In this study, real-time PCR was used to analyze the expression of the quorum sensing gene (*lasI*) of *Pseudomonas aeruginosa* under the exposure of thymoquinone. To carry out this experiment, organisms were allowed to grow separately under different concentrations (5 and 10 μg/mL) of thymoquinone followed by incubation at 37 °C for 48 h. A control set was prepared in which cells were grown in the absence of thymoquinone. Thereafter, trizol was used to extract mRNA from both thymoquinone treated and untreated bacterial cells. The purity of the collected mRNA was determined by using a spectrophotometer. Next, the entire RNA from each sample was allowed to get transcribed into the first strand of cDNA using the reverse transcriptase enzyme of M-MLV (Takara). Then, the same amount of cDNA (1 mg) was taken from each experimental set and subsequently real-time-amplified using the SYBR Premix kit (Applied Biosystems). In the negative control, a reaction mixture was prepared in which cDNA was not used. The expression level of the quorum sensing gene (*lasI*) was calculated under the influence of thymoquinone using the comparative threshold cycle method ($2^{-\Delta\Delta Ct}$) with *16S rRNA* as the control gene. The gene-specific primers along with the sequences used in this study are presented in Table 1.

Molecular docking analysis of thymoquinone with quorum sensing (QS) protein of *Pseudomonas aeruginosa*

A molecular docking study was performed to study the antibiofilm activity of thymoquinone against the quorum sensing protein (LasI) of the biofilm-forming organism, *Pseudomonas aeruginosa*. The crystal structure of the protein namely LasI (PDB ID: 1RO5) was retrieved from the Protein Data Bank (Berman et al. 2000). The PubChem (Wang et al. 2009) was used to download the 3D structure of thymoquinone (CID: 10281) in sdf format which was then converted into PDB format using UCSF Chimera (Pettersen et al. 2004). The protein and ligand were first energy-minimized prior to docking. Autodock Tools were used to add the polar hydrogens and Gastegier partial charges to the file following the docking using AutoDock Vina (Trott and Olson 2010) and AutoDock Tools (ADT) (Sanner 1999). To validate the binding affinity and stability of docked protein-ligand complex, energy minimization was further performed using GROMACS (Pronk et al. 2013). The interaction between the protein-ligand complexes was then visualized by generating 2D representation using the LigPlot programme and the 3D structures were visualized using PyMOL (DeLano 2002).

Pyocyanin quantification assay

The quantification assay of pyocyanin was carried out by following the protocol as described previously (Essar et al. 1990; Das et al. 2017). To perform the assay, *Pseudomonas aeruginosa* was inoculated in autoclaved LB growth media (5 mL) augmented with sub-MIC doses (5 and 10 μg/mL) of thymoquinone and incubated for 2 days at 37 °C. Additionally, a control set was also assigned in which a similar number of cells were grown without being exposed to thymoquinone. After the incubation, all the microbial cultures were centrifuged at 3000×*g* for 10 min to get the cell-free supernatant. Subsequently, 5 mL cell-free supernatant was extracted with 3 mL of chloroform and further re-extracted with 1 mL of 0.2 (M) HCl. The extracted pyocyanin showed yellow-to-pink-coloured solution. Then, the concentration of the coloured solution was determined by measuring the absorbance at a wavelength of 520 nm.

Cytotoxicity assay by MTT method

A549 cells were grown in 96-well plates in 0.2 mL of Dulbecco's Modified Eagle Medium (DMEM) with 10% Fetal bovine serum (FBS) and 1% antibiotics (Pen-Strep) and were cultured at 5% CO_2 at 37 °C for 24 h. The growth media in the wells were replaced after 24 h with fresh medium containing different concentrations of the antibiofilm agents individually or in combinations and incubated for another 24 h. After that, the medium was replaced with a fresh media containing 100 μL of 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) and incubated further for 4 h. Thereafter, the remaining unreduced MTT was removed from each well and 200 μL of DMSO was added thereafter to dissolve the produced MTT formazan crystals. Finally, the extent of the colour generated in each well was measured at 590 nm in a microplate reader.

Statistical Analysis

In the current manuscript, all the experiments were repeated three times. Significance test was carried out and the result of the same was incorporated in the form of P value < 0.05 (*), P value < 0.01 (**), and P value < 0.001 (***) in contrast to control. P value beyond 0.05 was defined as N.S. (no statistical difference).

Table 1 List of primer sequences used in real-time PCR

Name of bacteria used	Name of gene	Primer sequences used
Pseudomonas aeruginosa (accession number MTCC 424)	*lasI*	Forward 5′ CGT GCT CAA GTG TTC AAG G3′
		Reverse 5′TAC AGT CGG AAA AGC CCA G3′
	16S rRNA	Forward 5′ACT CCT ACG GGA GGC AGC GT3′
		Reverse 5′TAT TAC CGC GGC TGC TGG C3′

Results

Thymoquinone exhibited efficient antimicrobial activity against *Pseudomonas aeruginosa*

In the present study, the antimicrobial effect of thymoquinone has been examined against the test organism, *Pseudomonas aeruginosa*. To determine the same, various concentrations of thymoquinone were separately added to the wells of microbial lawn on LB agar plates and subsequently incubated at 37 °C for 2 days. After the incubation, prominent clear zones were noticed around the wells (Fig. 1a). The diameter of the clear zone around the wells was also measured wherein it was noticed that with the increase of thymoquinone concentrations, the diameter of the clear zone also increased (Fig. 1b). Thus, the result indicated a promising antibacterial activity of thymoquinone against *Pseudomonas aeruginosa*. To accurately measure the antimicrobial activity of thymoquinone against *Pseudomonas aeruginosa*, the broth dilution system (BDS) assay was followed (Das et al. 2016a). The result showed that with an increase in thymoquinone concentration, the growth of the tested organism reduced considerably (Fig. 1c). The result of the same indicated that thymoquinone at a concentration of 20 μg/mL prevented the visible growth of the mentioned bacteria, and hence, this concentration (20 μg/mL) of thymoquinone could be considered the minimum inhibitory concentration (MIC) against *Pseudomonas aeruginosa* (Fig. 1c). Moreover, the turbidity of all the growth media was quantified by recording their optical density (OD) at 600 nm. The result showed that with the gradual rise in the thymoquinone concentration, the extent of its turbidity in the growth media significantly decreased putting forth its promising antimicrobial activity against the bacteria (Fig. 1d). To gain further confidence, the minimum bactericidal concentration (MBC) of thymoquinone was determined and the result of the same was found to be 50 μg/mL against *Pseudomonas aeruginosa* (Fig. 1e). The result demonstrated that thymoquinone at a concentration of 50 μg/mL was able to kill all the microorganisms in the growth media. The antimicrobial activity of DMSO (6 μL) was also determined against *Pseudomonas aeruginosa* wherein we noticed that DMSO did not show any antimicrobial activity against the test organism (data not shown). Thus, the results revealed that thymoquinone exhibited substantial antimicrobial property against *Pseudomonas aeruginosa*.

Thymoquinone exhibited promising antibiofilm activity against *Pseudomonas aeruginosa*

To test the biofilm-inhibiting property of thymoquinone against *Pseudomonas aeruginosa*, a similar number of cells (1×10^7 CFU/mL) were separately added into different tubes carrying an equal volume (5 mL) of autoclaved LB growth media. After that, microorganisms in the growth media were further challenged with various sub-MIC concentrations (5 and 10 μg/mL) of thymoquinone. In the control set, a similar number of organisms were grown without being exposed to thymoquinone. All the growth media were then incubated for 2 days at 37 °C and subjected to the estimation of biofilm formation through CV assay. The result of the CV assay indicated that thymoquinone treatment was found to show a considerable reduction in the degree of biofilm formation by *Pseudomonas aeruginosa* (Fig. 2a). The result also revealed that thymoquinone at the concentrations of 5 and 10 μg/mL exhibited ~ 52 and ~ 62% diminutions in biofilm formation, respectively, in contrast to control in which the cells were grown without the exposure of thymoquinone (Fig. 2a). To gain further confidence, the degree of biofilm formation under different conditions was indirectly quantified by measuring the total biofilm protein (Tribedi and Sil 2014). The result showed that the protein recovery reduced considerably when the microorganisms were treated with thymoquinone in contrast to the control in which microorganisms were grown without being exposed to thymoquinone (Fig. 2b). Thus, the results revealed that thymoquinone was found to attenuate the biofilm-forming ability of the organism. To investigate further, efforts were put together to examine whether thymoquinone could interfere with the microbial colonization as it happens to be the first stage of biofilm formation (Chakraborty et al. 2020). The result showed that *Pseudomonas aeruginosa* in the absence of thymoquinone could efficiently colonize over the coverslips (Fig, 2c). However, the same organism showed a reduced degree of colonization under the influence of thymoquinone (Fig. 2c). To gain further confidence, a relation was established among the concentrations of thymoquinone, the extent of microbial colonization, and the biofilm inhibition wherein we observed that an increase in the concentrations of thymoquinone was found to decrease the microbial colonization that resulted in the inhibition of microbial biofilm formation (Supplementary Figure 1). Thus, the results indicated that the sub-MIC concentrations of thymoquinone were found to inhibit the microbial biofilm development considerably.

Thymoquinone exhibited a considerable accumulation of ROS in *Pseudomonas aeruginosa* that resulted in the inhibition of microbial biofilm formation

To examine the effect of thymoquinone on ROS generation, microorganisms were allowed to grow both in the presence and absence of thymoquinone under similar conditions. Intracellular ROS accumulation was then compared between thymoquinone-treated and thymoquinone-untreated microorganisms. The result showed that there was a significant variation in the ROS accumulation between the thymoquinone-treated and thymoquinone-

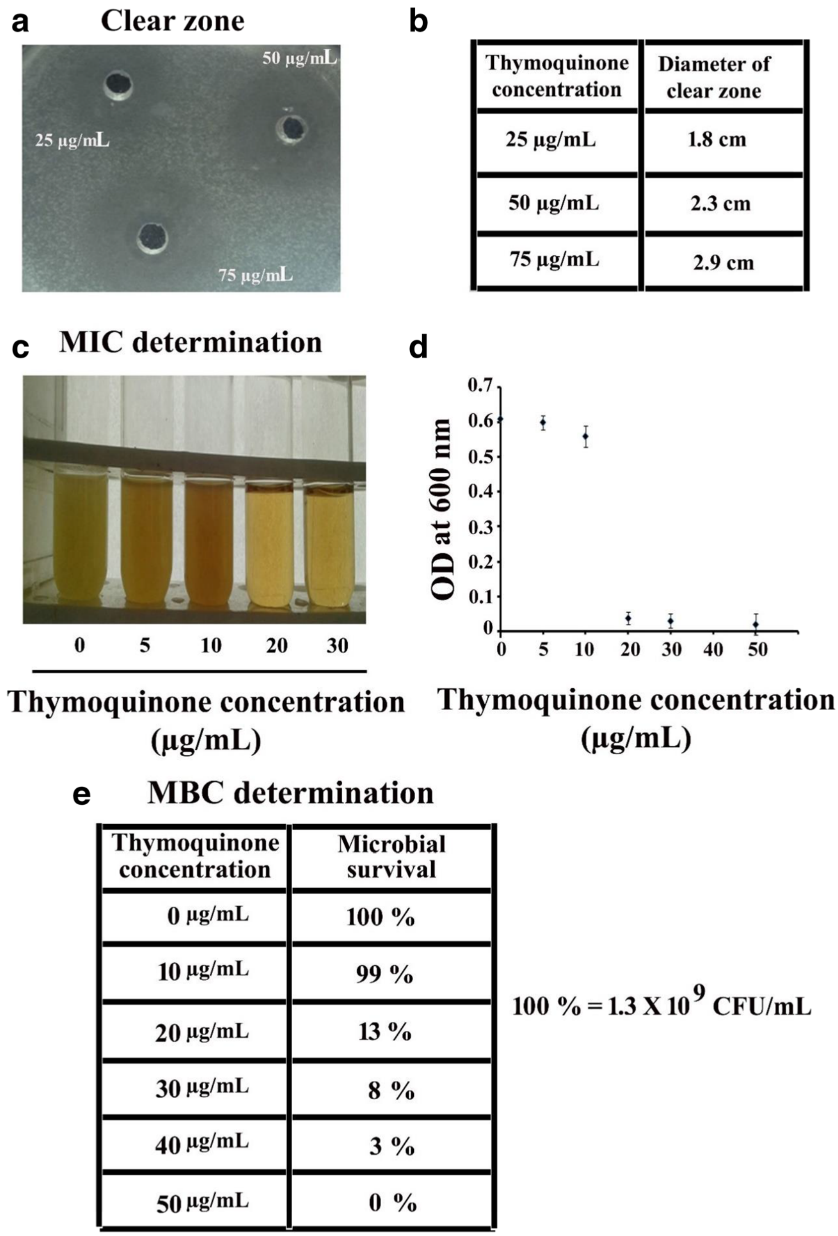


Fig. 1 Thymoquinone showed considerable antimicrobial activity against *Pseudomonas aeruginosa*. **a** Agar diffusion assay. Different concentrations of thymoquinone were added to the lawn of *P. aeruginosa* prepared on the LB agar plate to examine its antimicrobial activity. **b** Measurement of the diameter of the clear zone. The antimicrobial activity of thymoquinone was assessed by measuring the diameter of the clear zone around the well. **c** MIC determination. A similar number of *P. aeruginosa* were separately added into various test tubes containing sterile LB. Different concentrations of thymoquinone were separately added into each tube. All the growth media were then incubated for 48 h at 37 °C. Afterward, the extent of turbidity was analyzed wherein the tube with no visible turbidity refers to the MIC dose of thymoquinone. **d** Determination of microbial growth. The microbial growth of each growth media was separately measured by recording their OD at 600 nm. **e** MBC determination. A similar number of bacterial cells were inoculated into different test tubes having 5 mL of sterile LB growth media. Then, an increased concentration of thymoquinone was separately added to each growth media and incubated for 48 h at 37 °C. After that, the microbial viability was measured by determining the colony-forming unit. Experiments were conducted in triplicate and the result represented the average of three experiments

untreated cells (Fig. 3a). The result further showed that with the increase of thymoquinone concentrations, the ROS accumulation also increased significantly (Fig. 3a). To examine the effect of ROS accumulation on biofilm inhibition, various concentrations of ascorbic acid had been added to the cells which were already exposed to thymoquinone. The result showed that the tested concentrations of ascorbic acid drastically reduced the accumulation of ROS in those cells which were also treated with thymoquinone (Fig. 3a). To further reassure the effect of ROS on biofilm progress, several experiments were performed. In the first set, cells were treated with thymoquinone only (10 µg/mL). In the second set, various concentrations (10 and 20 µg/mL) of ascorbic acid were added to the cells which were treated with thymoquinone (10 µg/mL) as well. In the final set, cells were allowed to grow without being exposed to thymoquinone and ascorbic acid. The result revealed that thymoquinone treatment significantly inhibited microbial biofilm formation (Fig. 3b). However, the same organism restored the ability to develop biofilm under the influence of ascorbic acid (Fig. 3b). The result demonstrated that thymoquinone exposure was found to increase the ROS accumulation in the bacteria that resulted in the inhibition of microbial biofilm formation.

The accumulation of ROS exhibited moderate antibacterial activity

To examine the cellular accumulation of ROS on microbial viability, several experiments were conducted wherein in the

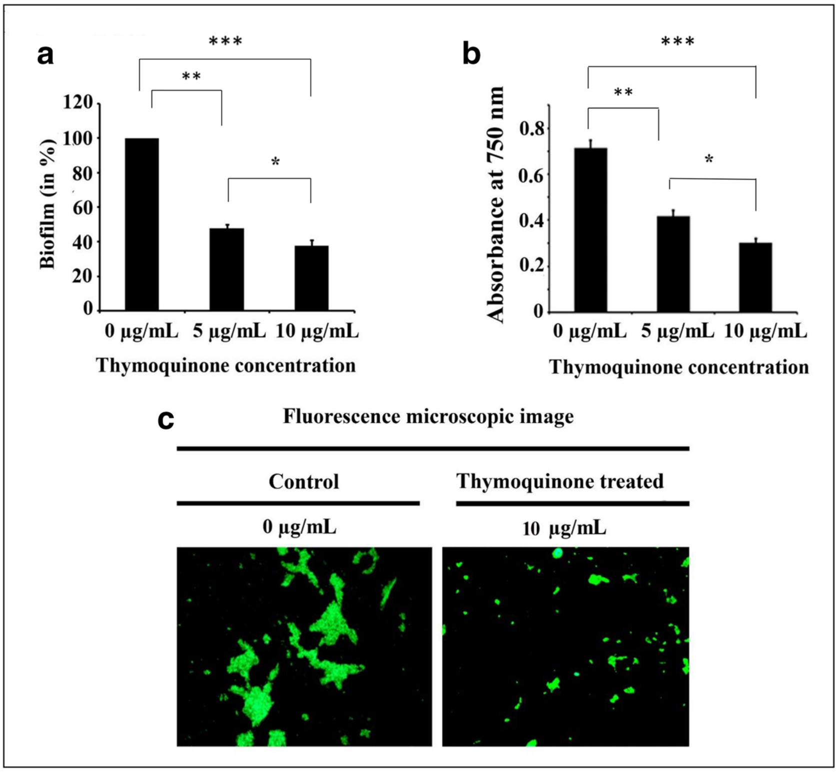


Fig. 2 Thymoquinone showed considerable antibiofilm activity against *Pseudomonas aeruginosa*. **a** CV assay. A similar number of *Pseudomonas aeruginosa* were separately grown both in the absence and presence of sub-MIC doses of thymoquinone and subjected to the measurement of biofilm formation by performing CV assay as mentioned in the materials and methods. **b** Total biofilm protein count profile. A similar number of *Pseudomonas aeruginosa* were grown both in the absence and presence of sub-MIC doses of thymoquinone for various length of time. Microbial adherence over the surface of the tube was counted by measuring the total protein. Experiments were conducted in triplicate. Error bars were used to designate the standard deviation (± SD). The P value < 0.05 (marked with *), < 0.01 (marked with **), and < 0.001 (marked with ***) were considered to measure the significant difference among the results. **c** Fluorescence microscopic image analysis. Microbial colonization to glass coverslips under the presence and absence of thymoquinone was examined by observing the acridine orange–stained cells under a fluorescence microscope. The figure is representative of images captured from 20 different fields and three independent experiments

first set, cells were allowed to grow with the MIC dose of thymoquinone (20 µg/mL). In the second set, different concentrations of ascorbic acid were added to the cells that were previously treated with the MIC dose (20 µg/mL) of thymoquinone as well. In the third set, cells were grown neither in the presence of thymoquinone nor ascorbic acid. All the growth media were then incubated for 48 h at 37 °C and subjected to the measurement of cell growth by measuring the OD at 600 nm. The result showed that microorganisms that were not treated with thymoquinone and ascorbic acid experienced the highest microbial growth (Supplementary Figure 2). However, cells that were treated with thymoquinone alone showed the least microbial growth as almost all the microorganisms were killed by thymoquinone (Supplementary Figure 2). The result also confirmed that ascorbic acid treatment restored the microbial growth of cells that were treated with the MIC dose of thymoquinone (20 µg/mL) too (Supplementary Figure 2). The result indicated that thymoquinone treatment was found to generate ROS in cells that might act as a causative agent for the death of the organisms. Existing literature revealed that the accumulated ROS could damage the cellular DNA considerably (Dwivedi et al. 2014). Therefore, in this study, efforts were given to examine the effect of ROS on cellular DNA integrity. To do the test, a similar number of cells (1×10^8 CFU/mL) were allowed to grow in sterile LB supplemented with various concentrations (10 and 15 µg/mL) of thymoquinone. In the control set, a similar number (1×10^8 CFU/mL) of cells were grown without being exposed to thymoquinone. After the stipulated period of incubation got over, every growth media including control was centrifuged. After the centrifugation, the cell pellets and supernatants were separately collected from both thymoquinone treated and untreated culture media. Afterward, the genomic DNA was isolated from a similar number of cells (1×10^8 CFU/mL) collected from the obtained cell pellets. The DNA collected from the respective cell pellets and the DNA present in the supernatants (if any) were run through agarose gel electrophoresis. The result showed that the cells that were not treated with thymoquinone showed an intense genomic DNA band in pellet whereas no DNA band

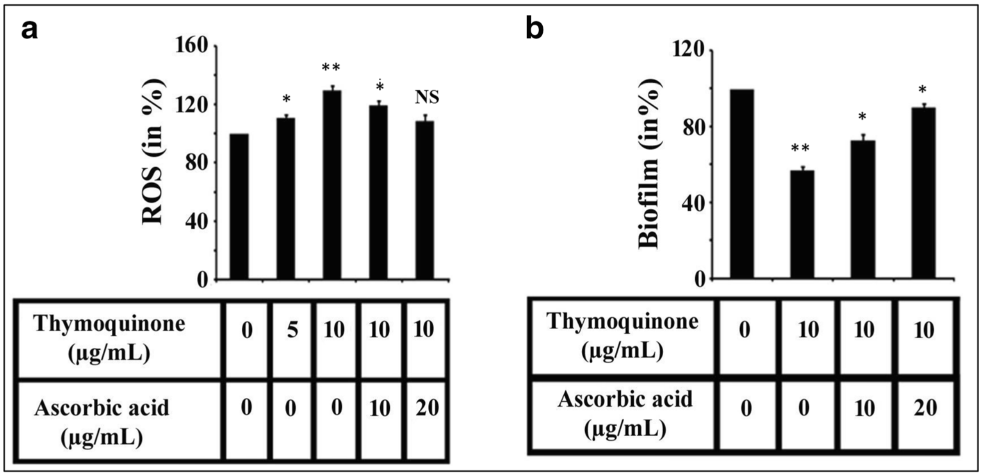


Fig. 3 ROS accumulation showed a considerable reduction in biofilm formation by *Pseudomonas aeruginosa*. **a** ROS measurement assay. Microorganisms were grown in LB augmented with varied concentrations of thymoquinone and ascorbic acid. All the growth media were then incubated for 2 days at 37 °C. Thereafter, an equal number of cells were independently recovered from each growth media. The intracellular ROS was separately measured in each tube by following the protocol as mentioned in the "Materials and methods". **b** Ascorbic acid re-established the biofilm-forming ability of microorganism that was treated with thymoquinone. An equal number of cells were allowed to grow in LB supplemented with various concentrations of thymoquinone and ascorbic acid. All the growth media were then incubated for 2 days at 37 °C. Then, planktonic cells were discarded from all growth media, and biofilm assay was performed by using a CV solution as described before. Experiments were conducted in triplicate. Error bars were used to designate the standard deviation (± SD). The P values < 0.05 (marked with *) and < 0.01 (marked with **) were considered to measure the significant difference among the results. N.S. indicated no statistical difference exists between the results

was observed in the supernatant of the same cells (Fig. 4). However, a mild-to-moderate decrease in the intensity of the genomic DNA band was noticed in the pellet of thymoquinone-treated cells (Fig. 4). It was further observed that some prominent DNA bands were also spotted in the supernatant of thymoquinone-treated (15 µg/mL) cells (Fig. 4). The results indicated that the treatment of thymoquinone was found to accumulate cellular ROS that resulted in the fragmentation of genomic DNA leading to the death of the microbial cells.

Thymoquinone exhibited considerable inhibition in the quorum sensing property of *Pseudomonas aeruginosa*

Quorum sensing, a density-dependent microbial communication event, has been found to influence several microbiological activities including swarming motility, biofilm formation, and the secretion of pyocyanin (Bordi and de Bentzmann 2011; Gupta et al. 2016; Das et al. 2016a; Chakraborty et al. 2018a). To understand the effect of thymoquinone on the swarming motility of *Pseudomonas aeruginosa*, a similar number of thymoquinone-treated and thymoquinone-untreated cells were spot-inoculated on the centre of the growth media and then incubated at 37 °C for 48 h. The result showed that *Pseudomonas aeruginosa* which was not treated with thymoquinone showed the greater colony diameter compared to the colony diameter of the cells that were treated with different concentrations of thymoquinone (Supplementary Figure 3). Thus, the result demonstrated that thymoquinone inhibited the swarming motility of the organism, indicating a possible influence of thymoquinone

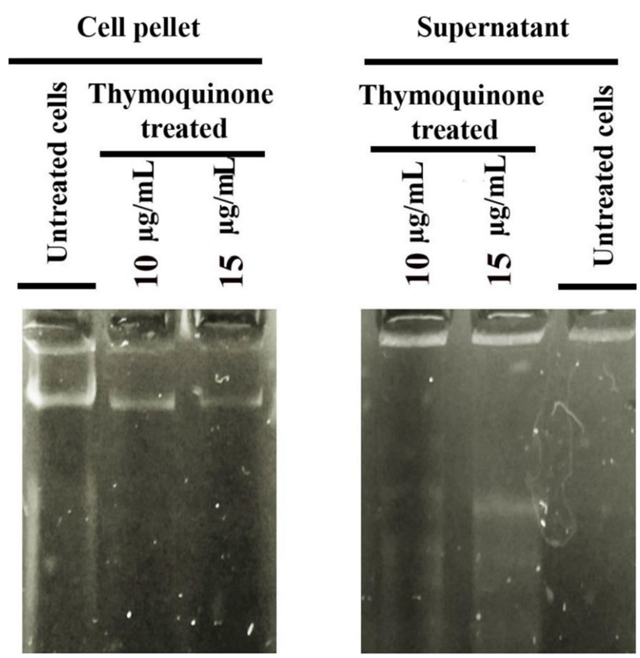


Fig. 4 Thymoquinone showed considerable interference with the genomic DNA integrity of *Pseudomonas aeruginosa*. A similar number of microorganisms were allowed to grow either in the presence or absence of thymoquinone for the required length of time. After the incubation, DNA was collected separately from the cell pellet and supernatant of both thymoquinone-treated and thymoquinone-untreated growth media. The collected DNA from a different source was then run through the agarose gel

over quorum sensing property of the organism. To examine the effect of thymoquinone on the quorum sensing property of the organism, the variation in the expression of quorum sensing gene (*lasI*) of *Pseudomonas aeruginosa* was measured both in the presence and absence of thymoquinone. The result revealed that the treatment of thymoquinone reduced the expression of quorum sensing–associated gene (*lasI*) in *Pseudomonas aeruginosa* as compared to the control in which a similar number of cells were grown in the absence of thymoquinone (Fig. 5a). To gain further confidence, a control experiment was performed in which the expression level of a gene (*16S rRNA*) unrelated to quorum sensing was measured both in the presence and absence of thymoquinone. The result demonstrated that no considerable difference was noticed between thymoquinone-exposed and thymoquinone-unexposed growth media in the context of the expression of *16S rRNA* gene (data not shown). Therefore, the result indicated that thymoquinone could selectively affect the expression of quorum sensing–associated gene (*lasI*) only (Fig. 5a). Thus, thymoquinone was found to interfere with the quorum sensing property of *Pseudomonas aeruginosa* that resulted in the inhibition of microbial biofilm formation. Besides, an in silico analysis was carried out to test whether thymoquinone could bind with the quorum sensing protein (LasI) of *Pseudomonas aeruginosa*. The result showed that the LasI protein exhibited a high binding affinity for thymoquinone (Fig. 5b). To check the stability of the docked complex, further energy minimization was done for the protein-ligand docked complex. The result showed that LasI protein could show a promising binding affinity for thymoquinone (− 5.9 kcal/mol) suggesting the considerable influence of thymoquinone on the quorum sensing property of the organism. It was also reported that *Pseudomonas aeruginosa* often involves quorum sensing for the production of pyocyanin (Das et al. 2016a). Hence, efforts were made to examine whether thymoquinone could inhibit the production of pyocyanin from *Pseudomonas aeruginosa*. To do the same, the microbial secretion of pyocyanin was compared between thymoquinone-treated and thymoquinone-untreated microorganisms. It was observed that thymoquinone-treated cells exhibited a lower level of pyocyanin secretion than the cells that were not treated with thymoquinone (Supplementary Figure 4). Thus, the results demonstrated that thymoquinone could interfere with the quorum sensing property of the organism that resulted in the inhibition of swarming motility, biofilm development, and pyocyanin production.

Combinations of thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles showed efficient antibiofilm activity against *Pseudomonas aeruginosa*

Existing literature reported that the combinatorial application of antibiofilm molecules was found to work more efficiently than individual ones in the context of biofilm attenuation (Dosler and Karaaslan 2014). In this regard, in addition to thymoquinone, two other molecules namely tryptophan (a natural amino acid) and tetrazine-capped silver nanoparticles (a synthetic molecule) had been taken into consideration for the combinatorial studies. The result showed that thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles were found to attenuate biofilm formation considerably when they were applied separately on *Pseudomonas aeruginosa* (Fig. 6a). However, the combinations (thymoquinone and tetrazine-capped silver nanoparticles; tryptophan and tetrazine-capped silver nanoparticles; thymoquinone and tryptophan; thymoquinone, tetrazine-capped silver nanoparticles, and tryptophan) were found to show an increased antibiofilm activity in comparison to their individual influences (Fig. 6a). It was also noted that the highest inhibition in biofilm development was observed when thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles were applied together on *Pseudomonas aeruginosa* (Fig. 6a). To gain further confidence, EPS secretion was measured under various conditions as EPS secretion has been considered an effective indicator of biofilm formation by microorganisms. To measure the EPS secretion, a similar number of cells of *Pseudomonas aeruginosa* were treated with different combinations of the mentioned antibiofilm molecules and the extent of EPS secretions was measured accordingly in each experimental condition. The result showed that the pattern of microbial EPS secretion varied when the cells were separately exposed to thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles (Fig. 6b). However, the highest inhibition in EPS secretion was noticed when thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles were applied together on *Pseudomonas aeruginosa* (Fig. 6b). Thus, the results showed that the combinatorial approaches of antibiofilm molecules were found to work more efficiently than the individual compounds in the context of biofilm attenuation.

Combinations of thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles efficiently inhibited microbial colonization

Since microbial colonization has been considered an important step of biofilm development (Chakraborty et al. 2020), in this study, efforts were made to examine whether the combinatorial applications could inhibit the microbial colonization of *P. aeruginosa*. In this regard, we observed that microorganisms not treated with thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles showed densely associated microbial clusters which suggested substantial biofilm development on the surface (Supplementary Figure 5). However, it was also observed that the tested combinations of the mentioned antibiofilm compounds were found to inhibit the microbial colonization effectively in contrast to the control in which cells were allowed to colonize in the absence of any of the mentioned antibiofilm agents (Supplementary

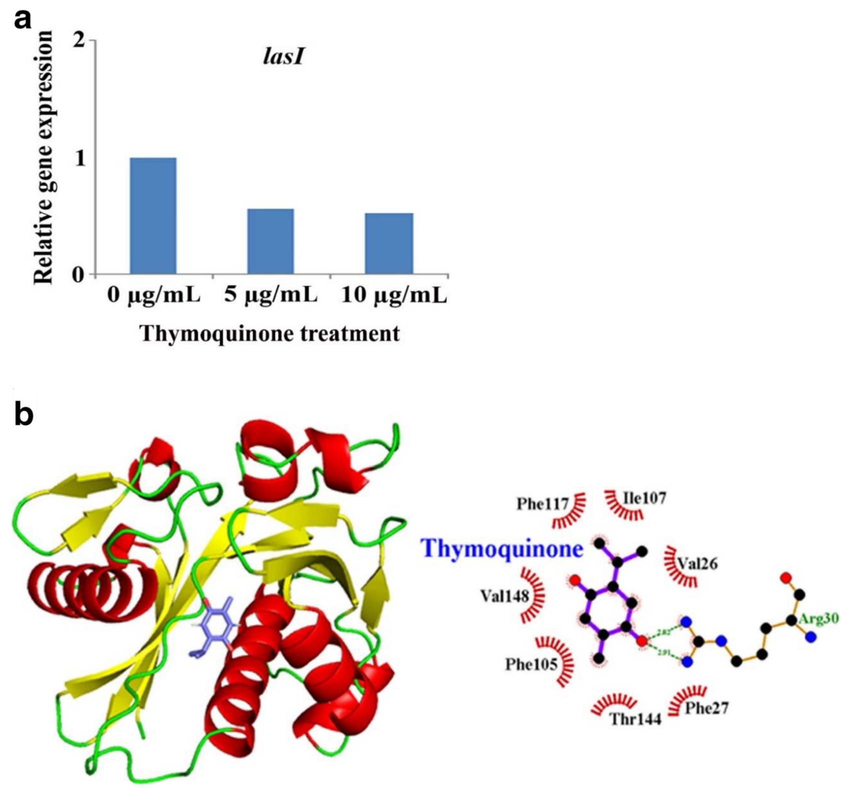


Fig. 5 Thymoquinone inhibited the quorum sensing–linked gene expression of *P. aeruginosa*. **a** Real-time PCR analysis. Cells were allowed to grow both in the presence and absence of thymoquinone at 37 °C for 2 days. Thereafter, RNA was isolated from both thymoquinone-treated and thymoquinone-untreated organisms, and the expression of the quorum sensing gene (*lasI*) was analyzed by real-time PCR analysis. **b** Molecular docking and interaction analysis of thymoquinone with quorum sensing–related protein (LasI) of *P. aeruginosa*. Cartoon structure of protein-ligand docked complex of LasI (PDB ID: 1RO5) and thymoquinone (left panel), LigPlot interaction analysis of thymoquinone with the active residues of LasI (right panel)

Figure 5). The highest inhibition in microbial colonization took place when the cells were allowed to colonize in the presence of thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles (Supplementary Figure 5). Thus, the combinatorial application of thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles were found to interfere with the microbial colonization considerably that resulted in the inhibition of microbial biofilm formation.

Combinations of thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles exhibited considerable accumulation of ROS that resulted in the inhibition of microbial biofilm development

To examine the effect of the combinations of the antibiofilm molecules on the cellular accumulation of ROS, a similar number of cells were exposed to several combinations of antibiofilm molecules and incubated under similar conditions. After the incubation, the cellular accumulation of ROS was measured accordingly in each experimental set. The result showed that thymoquinone and tetrazine-capped silver nanoparticles individually could increase ROS accumulation but tryptophan alone did not make any change in ROS accumulation status in the microorganism (Fig. 7). However, the result also showed that the prepared combinations were able to increase the ROS accumulation substantially in contrast to control sets in which the cells were grown in the absence of any of the mentioned antibiofilm molecules (Fig. 7). To establish the correlation (if any) between the accumulation of ROS and the development of microbial biofilm, a similar number of cells were grown in LB under different combinations of the tested antibiofilm agents and further challenged with ascorbic acid as it is a strong chelator of ROS. In the control set, cells were allowed to grow without any of the tested antibiofilm molecules and ascorbic acid. After that, all the growth media including the control were incubated at 37 °C for 2 days. Then, an equal number of microbial cells were separately collected from all the growth media and the microbial accumulation of ROS and the extent of biofilm formation by the microorganisms were examined accordingly. The result showed that in most of the cases, ROS accumulation established an inversely proportional relationship with the microbial biofilm development (Fig. 8). It was also noticed that whenever ROS accumulation went up, the extent of microbial biofilm development went down immediately (Fig. 8). Moreover, the correlation coefficient (r) was found to be − 0.91 between ROS accumulation and microbial biofilm development. Thus, the combinatorial approaches of antibiofilm molecules were found to inhibit biofilm formation by accumulating ROS in the microorganism.

Combinations of thymoquinone, tryptophan, and tetrazine-capped silver nanoparticles did not show considerable cytotoxicity

Cytotoxicity of the tested antibiofilm agents individually or in combination was assessed on A549, a human lung epithelial cell line. The result showed that there was a little variation

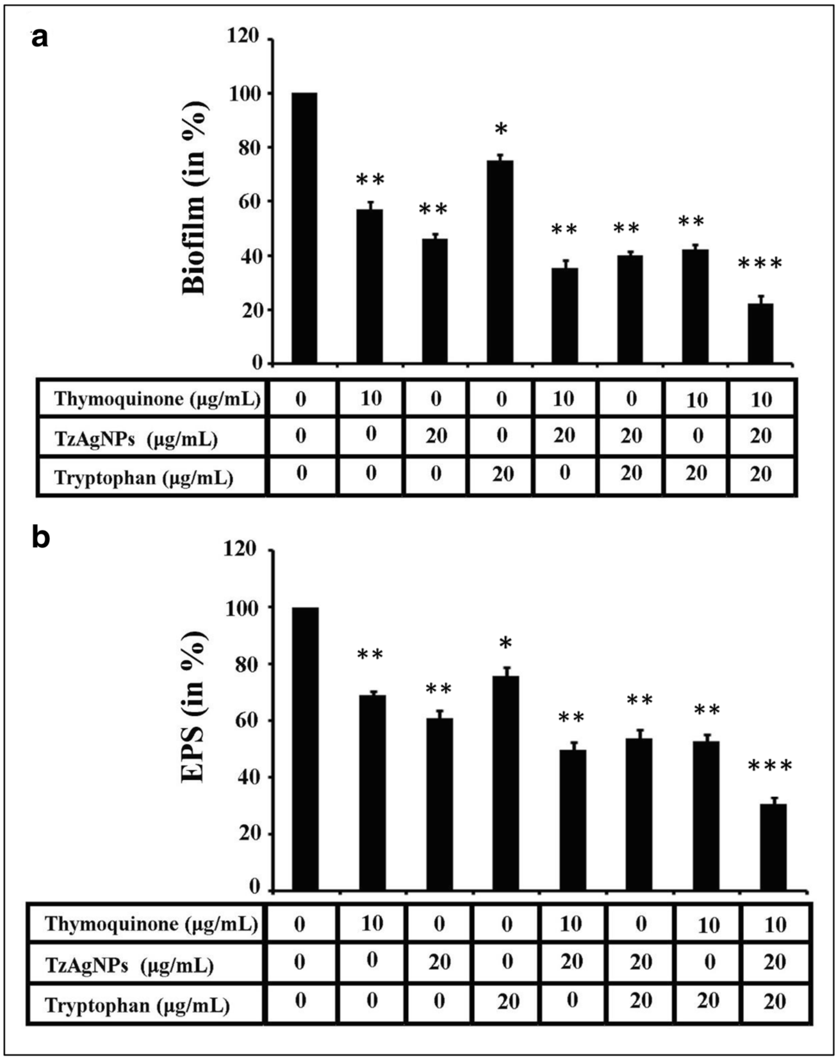
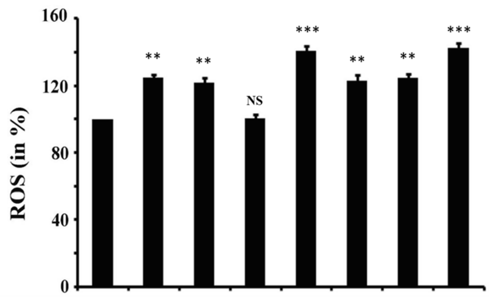


Fig. 6 Combinatorial application of antibiofilm molecules exhibited considerable biofilm inhibition against *Pseudomonas aeruginosa.* **a** Crystal violet (CV) staining assay. A similar number of bacterial cells were independently grown in LB supplemented with different combinations of TzAgNPs, thymoquinone, and tryptophan and incubated at 37 °C for 2 days. After the incubation, planktonic cells were removed and the adhered biofilm populations were separately stained with the CV solution. After that, the CV solution was discarded followed by the addition of 33% glacial acetic acid. The absorbance of the glacial acetic acid suspension in all tubes was recorded at 630 nm. **b** EPS measurement profile. An equal number of cells of *Pseudomonas aeruginosa* were allowed to grow individually in LB augmented with different combinations of TzAgNPs, thymoquinone, and tryptophan. The amount of EPS produced from the treated and untreated cells was measured by the phenol-sulphuric acid method. Experiments were conducted in triplicate. Error bars were used to designate the standard deviation (± SD). The *P* values < 0.05 (marked with *), < 0.01 (marked with **), and < 0.001 (marked with ***) were considered to measure the significant difference among the results

observed in the cell viability of A549 under various treatments (Fig. 9). It was observed that thymoquinone and tryptophan individually showed no significant cytotoxicity against A549 cells (Fig. 9). However, tetrazine-capped silver nanoparticles alone could show only a 7% reduction in cell viability in comparison to the control set where A549 cells were allowed to grow without being exposed to any of the tested antibiofilm molecules (Fig. 9). It was also observed that the combinations of the used antibiofilm agents either showed no noteworthy cytotoxicity or very little cytotoxicity against A549 cells (Fig. 9). Thus, the combined action of the antibiofilm molecules could be used as potential agents for the sustainable management of biofilm-associated infections caused by *Pseudomonas aeruginosa.*

Discussion

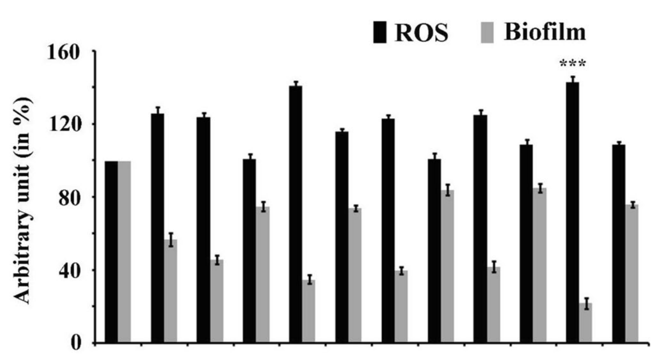
Microbial biofilm often exhibits antibiotic resistance and enhances the secretion of several virulence factors that play a key role in the promotion of microbial pathogenicity, thereby posing a serious threat to sustainable health management (Hall-Stoodley and Stoodley 2009). *Pseudomonas aeruginosa* has been found to cause several infections in



Thymoquinone (μg/mL)	0	10	0	0	10	0	10	10
TzAgNPs (μg/mL)	0	0	20	0	20	20	0	20
Tryptophan (μg/mL)	0	0	0	20	0	20	20	20

Fig. 7 Combinatorial application of antibiofilm molecules exhibited considerable inhibition in the accumulation of ROS in *Pseudomonas aeruginosa*. An equal number of bacterial cells were allowed to grow in LB media supplemented with different combinations of TzAgNPs, thymoquinone, and tryptophan. After the incubation got over, an equal number of microbial cells were separately collected from each media. Afterward, the intracellular accumulation of ROS was measured by following the procedure mentioned in the "Materials and methods". Experiments were conducted in triplicate. Error bars were used to designate the standard deviation (± SD). The *P* values < 0.01 (marked with **) and < 0.001 (marked with ***) were considered to measure the significant difference among the results. N.S. indicated no statistical difference noticed among the results

humans, exploiting biofilm formation (Gupta et al. 2016). In this regard, various natural and synthetic molecules have been explored by the scientific community across the globe to inhibit microbial biofilm formation (Gupta et al. 2016). Towards this direction, the antibiofilm activity of a natural molecule namely thymoquinone was examined against the test organism, *Pseudomonas aeruginosa*. Thymoquinone happens to be an active compound of the widely used medicinal plant, *Nigella sativa* (black seed) and has been known to inhibit the growth of various microorganisms (Chaieb et al. 2011). Several antimicrobial assays such as clear zone assay, turbidity measurement, and colony count were conducted to confirm the antimicrobial property of thymoquinone against *Pseudomonas aeruginosa.* To quantitatively determine this property of thymoquinone against the test organism, the minimum inhibitory concentration (MIC) of the compound was determined. The MIC of thymoquinone was found to be 20 μg/mL as this concentration happened to be the lowest concentration of thymoquinone that did not allow any visible microbial growth to appear after the required time of incubation. Hence, in this study, the sub-MIC concentrations (5 and 10 μg/mL) of thymoquinone were selected to examine its antibiofilm activity against *Pseudomonas aeruginosa* as the main objective of our study was not to kill the organisms rather inhibit the microbial biofilm formation. Several widely used techniques such as CV assay, protein count assay, and microscopic studies were performed to estimate the antibiofilm activity of thymoquinone against the mentioned organism. CV assay, being one of the most popular methods to quantitatively measure the microbial biofilm formation under various conditions (Das et al. 2016a; Chakraborty et al. 2020) had been chosen for the current study as well. Observations from the same indicated that thymoquinone could efficiently reduce the microbial biofilm formation by *Pseudomonas aeruginosa*. For further reassurance, the total biofilm protein was also measured to take into account the degree of microbial associations in the biofilm under different conditions as the traditional colony counting assay posed serious difficulties to extract the microorganisms adhered on the



Thymoquinone (μg/mL)	0	10	0	0	10	10	0	0	10	10	10	10
TzAgNPs (μg/mL)	0	0	20	0	20	20	20	20	0	0	20	20
Tryptophan (μg/mL)	0	0	0	20	0	0	20	20	20	20	20	20
Ascorbic acid (μg/mL)	0	0	0	0	0	20	0	20	0	20	0	20

Fig. 8 Combinatorial application of antibiofilm agents exhibited considerable inhibition in the biofilm formation through the accumulation of ROS in *Pseudomonas aeruginosa*. An equal number of bacterial cells were grown separately in LB media treated with various combinations of TzAgNPs, thymoquinone, tryptophan, and ascorbic acid. After the incubation, the extent of biofilm formation and the accumulation of ROS in microorganism was measured accordingly. Experiments were conducted in triplicate. Error bars were used to designate the standard deviation (± SD). The *P* value < 0.001 (marked with ***) was considered to measure a significant difference among the results

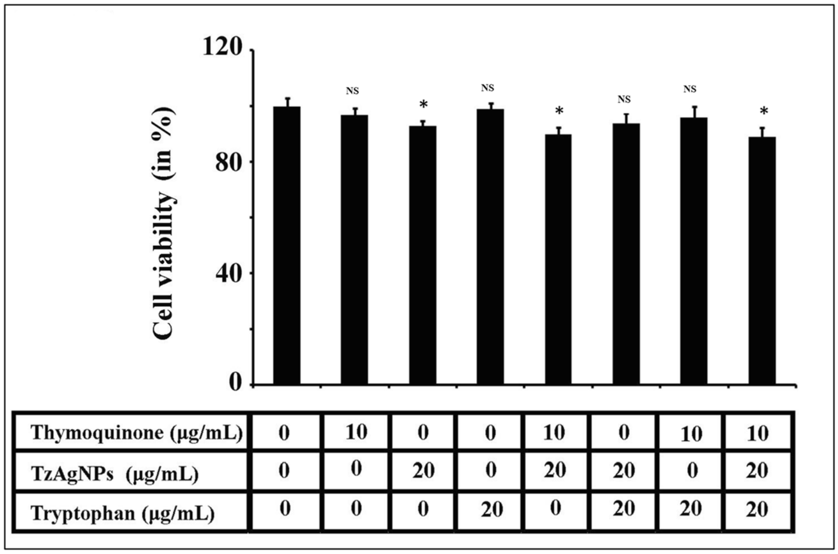


Fig. 9 Cytotoxicity assay. The cytotoxicity of thymoquinone, TzAgNPs, and tryptophan on A549 (a lung epithelial cells) was tested by following MTT assay as mentioned in the "Materials and methods" section. Experiments were conducted in triplicate. Error bars were used to designate the standard deviation (± SD). The *P* value < 0.05 (marked with *) was considered to measure the significant difference among the results. N.S. indicated no statistical difference spotted between the results

glass surfaces (Tribedi and Sil 2013; Tribedi and Sil 2014; Bhattacharyya et al. 2018). Literature reports stated that the protein recovery profile exhibited a directly proportional relationship with that of the degree of the microbial association on the glass surface (Tribedi and Sil 2014; Bhattacharyya et al. 2018). Additionally, the same was quite evident from our experimental results as well which indicated that the biofilm-forming ability of the microorganisms reduced under the influence of thymoquinone, thereby maintaining a similar profile as that of the CV assay. Since microbial colonization happens to be a key step towards biofilm formation, any agent interfering with microbial colonization could emerge as a potential antibiofilm molecule (Chakraborty et al. 2020). Hence, in this study, efforts were put together to examine the extent of microbial colonization under the influence of thymoquinone. We observed that the microorganism exhibited poor colonization under the exposure of thymoquinone leading to the reduced degree of biofilm formation. Taken together, the results demonstrated that thymoquinone exhibited considerable antibiofilm activity against *Pseudomonas aeruginosa*. Thereafter, to investigate the underlying mechanism of the same, microbial accumulation of ROS was measured between the thymoquinone-treated and thymoquinone-untreated cells as literature survey affirmed that ROS accumulation was able to reduce the microbial biofilm formation substantially (Chakraborty et al. 2018a). Towards this direction, we observed that thymoquinone could considerably increase the cellular accumulation of ROS. It was also noted that whenever ROS accumulation increased, microbial biofilm formation concomitantly decreased, suggesting a link of ROS accumulation behind microbial biofilm inhibition. Since several literatures reported that microbial accumulation of ROS exhibited considerable antimicrobial activity (Bendich et al. 1986; Dwyer et al. 2009), in this study, we examined whether the ROS generated by thymoquinone could exhibit any antimicrobial activity against *Pseudomonas aeruginosa*. In this connection, it was noted that ascorbic acid (an antioxidative agent) was able to neutralize the elevated ROS in cells treated with thymoquinone thereby restoring the microbial growth. In this regard, several other articles also reported that ROS accumulation could damage cellular biomolecules including DNA while exhibiting antimicrobial activity (Dwyer et al. 2009; Dwivedi et al. 2014). As expected, thymoquinone-untreated cells showed an integrated genomic DNA that did not diffuse out of the cells. However, during thymoquinone treatment, the genomic DNA got broken into small fragments and was released by the microorganism. Thus, this study brought forth observations in the same direction revealing that the treatment of thymoquinone showed a considerable accumulation of ROS in microorganism that resulted in the breakdown of the genomic DNA leading to cause cell death. Further explorations revealed that thymoquinone was found to inhibit the swarming motility and pyocyanin production from *Pseudomonas aeruginosa*, indicating that thymoquinone had the possibility to interfere with the quorum sensing property of *Pseudomonas aeruginosa* as it was documented in the literature that quorum sensing considerably influences the microbial swarming motility, biofilm formation, pyocyanin production, etc. (Gupta et al. 2016; Das et al. 2017). Quorum sensing is a phenomenon in which microorganisms communicate with each other through the synthesis of self-made autoinducers or pheromones (Gupta et al. 2016). *Pseudomonas aeruginosa* is seen to be capable of synthesizing autoinducers in the form of N-acyl homoserine lactone (Gupta et al. 2016). Autoinducers after reaching the threshold value control the expression of a gene connected to biofilm formation (Bordi and de Bentzmann 2011). It was reported in the literature that LasI protein functions in controlling the quorum sensing property of *Pseudomonas aeruginosa* (Zhu et al. 2004). The *lasI* gene codes for acyl-homoserine lactone synthase in the test bacterium that resulted in the production of autoinducer molecule {N-3-oxo-dodecanoyll-homoserine lactone (3O-C12-HSL)} (Kiratisin et al. 2002). The autoinducers trigger the transcriptional activator to regulate the expression of desired genes responsible for biofilm development (Kiratisin et al. 2002). In this study, we measured the expression of *lasI* gene of *Pseudomonas aeruginosa* under the presence and absence of thymoquinone. We observed that thymoquinone-treated cells showed a reduced degree of *lasI* gene expression in contrast to the cells not exposed to thymoquinone suggesting that thymoquinone could interfere with the quorum sensing property of *Pseudomonas aeruginosa*. To gain further confidence, molecular docking analysis was conducted in which we noticed that the compound thymoquinone could interact with the LasI protein considerably by forming different types of non-covalent and covalent bonds. Thus, the results demonstrated that the compound thymoquinone was able to interfere with the quorum sensing property of the organism that resulted in the inhibition of biofilm formation. In multiple reports, it was observed that the combinatorial applications of antibiofilm molecules could act more effectively than their individual presence in attenuating microbial biofilm (Das et al. 2016b; Das et al. 2017). Hence, in addition to thymoquinone, two other reported antibiofilm molecules, namely tetrazine-capped silver nanoparticles and tryptophan, had been taken into consideration for the noteworthy inhibition of biofilm formation by *Pseudomonas aeruginosa*. It was reported that 20 μg/mL concentration of tryptophan exhibited a substantial effect on microbial biofilm attenuation (Chakraborty et al. 2018b). In the same report, it was also documented that 20 μg/mL concentration of tryptophan showed no considerable cytotoxicity (Chakraborty et al. 2018b). Hence, this concentration of tryptophan was considered for this study. Moreover, for tetrazine-capped silver nanoparticles, we had observed that the compound with a concentration of 20 μg/mL exhibited the highest antibiofilm activity against *Pseudomonas aeruginosa* (Chakraborty et al. 2018a). Hence, this concentration of tetrazine-capped silver nanoparticles was selected for the present study. Several combinations were made among the mentioned antibiofilm molecules (thymoquinone and tetrazine-capped silver nanoparticles; tryptophan and tetrazine-capped silver nanoparticles; thymoquinone and tryptophan; thymoquinone, tetrazine-capped silver nanoparticles, and tryptophan) in which we observed that the highest inhibition in biofilm formation took place when the cells were treated with all the tested antibiofilm molecules. To investigate the underlying cause of biofilm inhibition under the exposure of the combinations of the antibiofilm molecules, it was observed that the combinatorial approach of the mentioned antibiofilm molecules could efficiently generate ROS that resulted in the substantial decrease of biofilm formation by the test organism. Moreover, for the safe application of the used antibiofilm molecules, the cytotoxicity effect of the tested combinations was pursued against the A549 cell line using MTT assay. MTT assay happens to be a colorimetric test relying on the functions of a cellular enzyme often being considered as the indicator for cell viability (Chakraborty et al. 2018b). The result indicated that the combinations of the tested antibiofilm molecules exhibited either very little toxicity or no toxicity and could be applied to manage the biofilm challenge in a cost-effective and sustainable fashion.

Conclusion

The inception of innovative agents that aid in the reduction of microbial biofilm paves a new path to fight against infections associated with biofilm. The effective combinations of the tested antibiofilm molecules were found to attenuate biofilm development comprehensively in contrast to their individual influence. The highest biofilm inhibition was observed when the cells were treated with thymoquinone, tetrazine-capped silver nanoparticles, and tryptophan together and hence could be used as potential agents for the sustainable management of biofilm-linked infections caused by *Pseudomonas aeruginosa*.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s12223-020-00841-1.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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